

Python Basics – AI Mentor Knowledge Base

Beginner-friendly reference material for the AI Mentor RAG system.

1. What is Python?

Python is a high-level, interpreted, general-purpose programming language. It is known for readable syntax and is widely used in web development, automation, data analysis, machine learning, artificial intelligence, and scientific computing.

2. Variables

A variable is a name that refers to a value. Python does not require explicit type declarations. Example: `name = 'Anila'` and `age = 19`. Variable names are case-sensitive and cannot start with a number.

3. Data Types

Common Python data types are `int` for integers, `float` for decimal numbers, `str` for text, `bool` for True or False, `list` for ordered mutable collections, `tuple` for ordered immutable collections, `set` for unique elements, and `dict` for key-value pairs.

4. Operators

Arithmetic operators include `+`, `-`, `*`, `/`, `%`, `//`, and `**`. Comparison operators include `==`, `!=`, `>`, `<`, `>=`, and `<=`. Logical operators are `and`, `or`, and `not`. Assignment operators include `=`, `+=`, `-=`, `*=`, and `/=`.

5. Conditional Statements

Python uses `if`, `elif`, and `else` for decision making. Example: `if marks >= 40: print('Pass') else: print('Fail')`.

6. Loops

A `for` loop iterates over a sequence such as a list or string. A `while` loop repeats while a condition is true. `break` stops a loop and `continue` skips the current iteration.

7. Functions

A function is a reusable block of code. Functions are defined with `def`. Example: `def add(a, b): return a + b`. Functions can accept parameters and return values.

8. Lists, Tuples, Sets and Dictionaries

A list is ordered and mutable, such as `[10, 20, 30]`. A tuple is ordered and immutable, such as `(10, 20, 30)`. A set stores unique elements, such as `{10, 20, 30}`. A dictionary stores key-value pairs, such as `{'name': 'Anila'}`.

9. Exception Handling

Exception handling helps programs deal with runtime errors. Python commonly uses `try`, `except`, `else`, and `finally`.

10. Object-Oriented Programming

Object-oriented programming uses classes and objects. A class is a blueprint and an object is an instance of a class. Important concepts include encapsulation, inheritance, polymorphism, and abstraction.

11. Python in Data Science and AI

Python has a large library ecosystem. NumPy is used for numerical computing, Pandas for data manipulation, Matplotlib for visualization, and scikit-learn for machine learning.

12. Beginner Learning Order

A useful learning order is: syntax and variables, data types, operators, conditions, loops, functions, collections, exception handling, modules, OOP, then libraries such as NumPy and Pandas.

13. AI Mentor Practice Questions

Examples include: What is Python? What is a variable? What are Python data types? What is the difference between a list and tuple? What is a function? What is a for loop? What is exception handling? What is a class and object? Why is Python used in AI?